Multi-Modal Measurements of Weather Effects and Link Quality Dynamics in LEO Satellite Networks

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Motivation & Contribution

- Rapid expansion of LEO Networks
  - Understanding link quality, and limiting and impacting factors, are important
- Our focus: weather effects (case study: Starlink)
- Novelty: Using professional instruments (from Chair of Climatology at TU Berlin)
- Research questions:
  - What hurts performance? Cloud coverage? Rain? Wind??
  - What is affected most? Throughput? Latency? Download? Upload?
- Methodological questions:
  - How to measure cloud coverage?
  - How to make sure diurnal patterns don‘t bias the data?
  - How to know to which satellite I am connected? (So I know which clouds are in that direction.)

Outline

1. **Introduction: LEO Satellites**
2. **Related Work**
3. **Methodology**
   a. Weather Instruments
   b. Network Measurements
   c. Image Classifier
   d. Detection of Connected Satellites
4. **Deployment**
5. **Results**
6. **Discussion / Future Work**

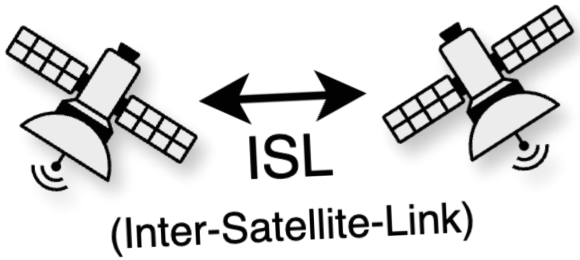
Introduction to Low-Earth-Orbit (LEO) Satellites

- Unlike traditional geostationary (GEO) satellites located at 35,786km, LEO satellites have a significantly closer orbit of **300km to 2000km**.
  - this proximity significantly **reduces latency**
  - offers new opportunities for **global connectivity** esp. in previously unserved areas

- Starlink e.g. also offers **Direct-to-Cell** to complement the current 5G infrastructures

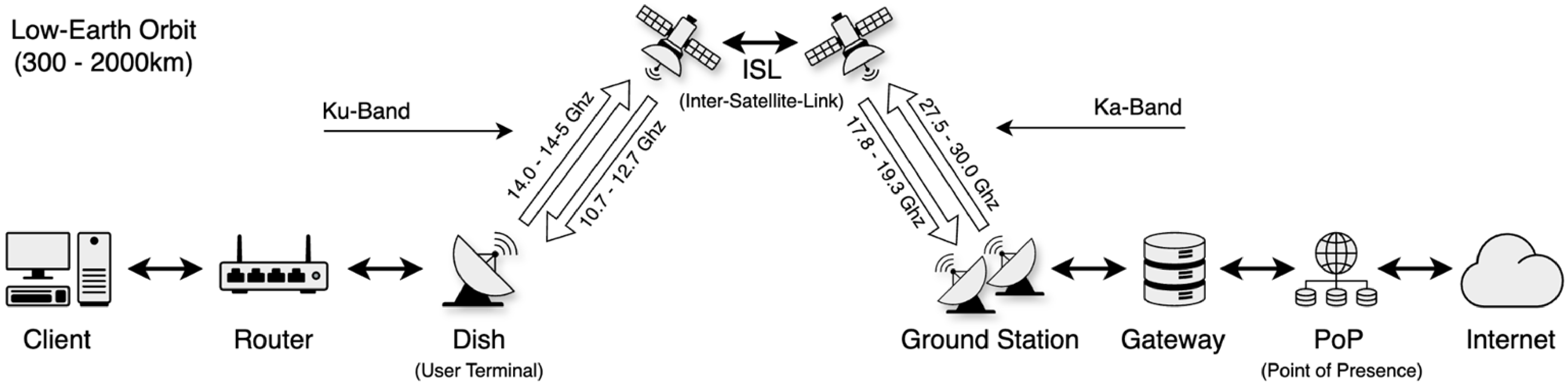
- LEO satellites are **sensitive to atmospheric influences** (e.g., temperature, rain, cloud cover, ionospheric activity), satellite motion, and local topography and obstructions [Ma+23].

- LEO satellites move at a speed of over **27,000 km/h** [BS19, p. 1]



Introduction to Starlink Infrastructure

- During our study (2025), Starlink had **7,207** active satellites in orbit (total 10,513)
- In our UDP-based performance tests with the current Starlink Standard Kit Gen. 3 (Rev. 4)
  - maximum download speeds of **490 MBit/s** and **150 MBit/s** upload speeds
  - minimum latency of **21ms**



Related Work

- Several works measured impact of **rain** [Kas+22; Ma+23]
- [Lan+24a]: **cloud cover** impacts download throughput by more **than 6 MBit/s**
- Also papers [Kas+22] and [Mic+22] which deal with end-user browser performance
  - a significant improvement over geostationary internet and first comparability with classical predominant DSL internet
  - up **to 146 MBit/s download link** and a **median RTT of 50ms**

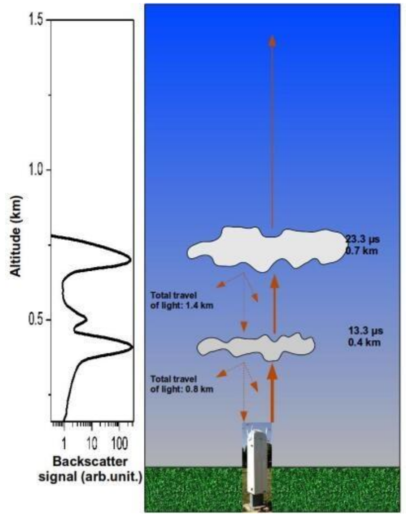
Our Approach: More Fine-Grained Weather Instruments

- In contast, our setup also **includes special cloud sensors:**
  - **Ceilometer**: ground-based remote-sensing device, uses light beam to measure height of clouds (e.g., via backscattering)
  - **Microwave radiometer**: measure humidity profiles, integrated water vapor (IWV*), **liquid water path** (LWP**)
  - For image-based estimation: **Astro Camera**
  - And others: **disdrometer*****, Eddy-Covariance system, etc.

*amount of **water vapor in vertical column** from the Earth's surface to top of the atmosphere

amount of **liquid water

***drop size distribution (DSD) and fall velocity



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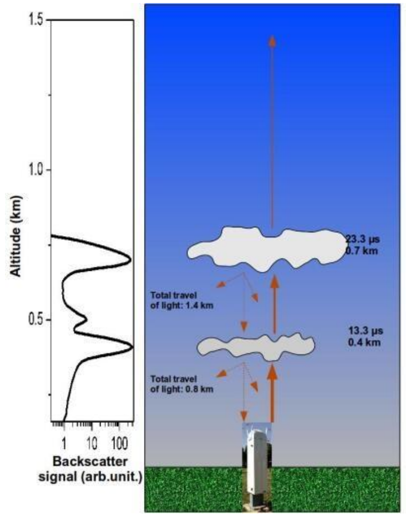

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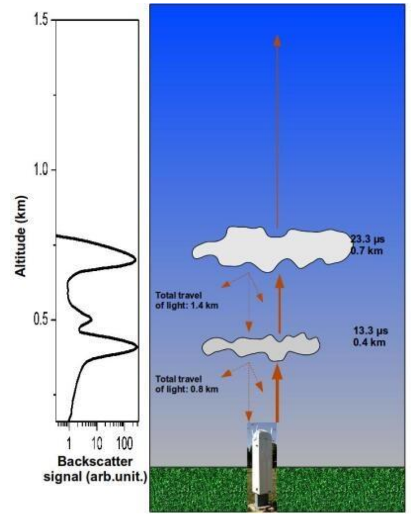
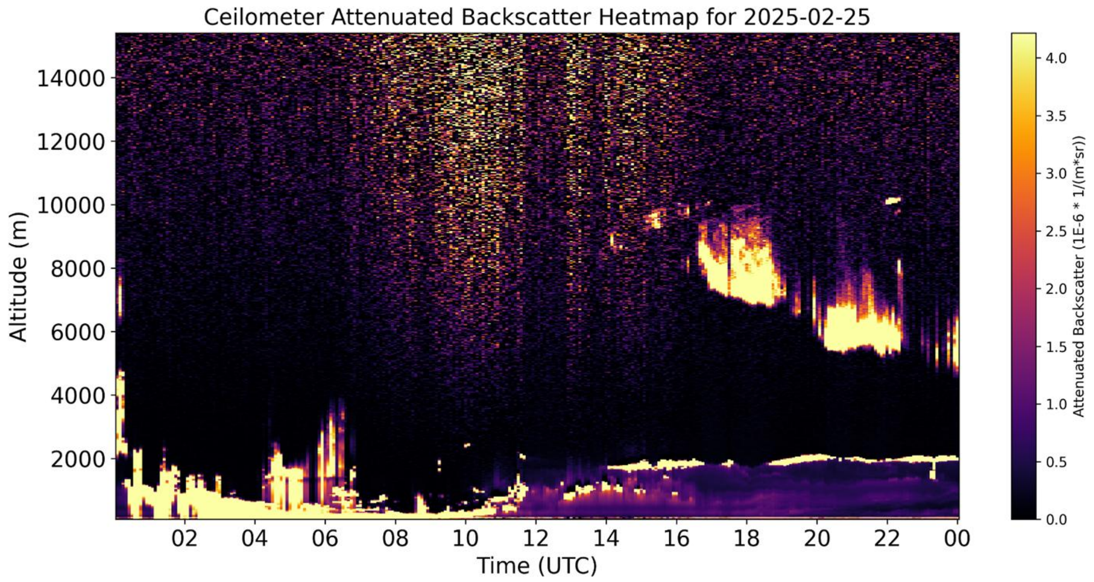


Our Measurements

- **112 high-quality** meteorological measurements
- Mostly in **1-min intervals**
- Weather instruments: details
    - Semi-professional 7-in-1 weather station
    - Microwave-Radiometer (RPG-HATPRO-G5)
    - Ceilometer (Lufft CHM15k)
    - Disdrometer (Thies Clima LNM)
    - Eddy-Covariance-System
    - Sky Camera (ZWO AstroCam)



Ceilometer Attenuated Backscatter

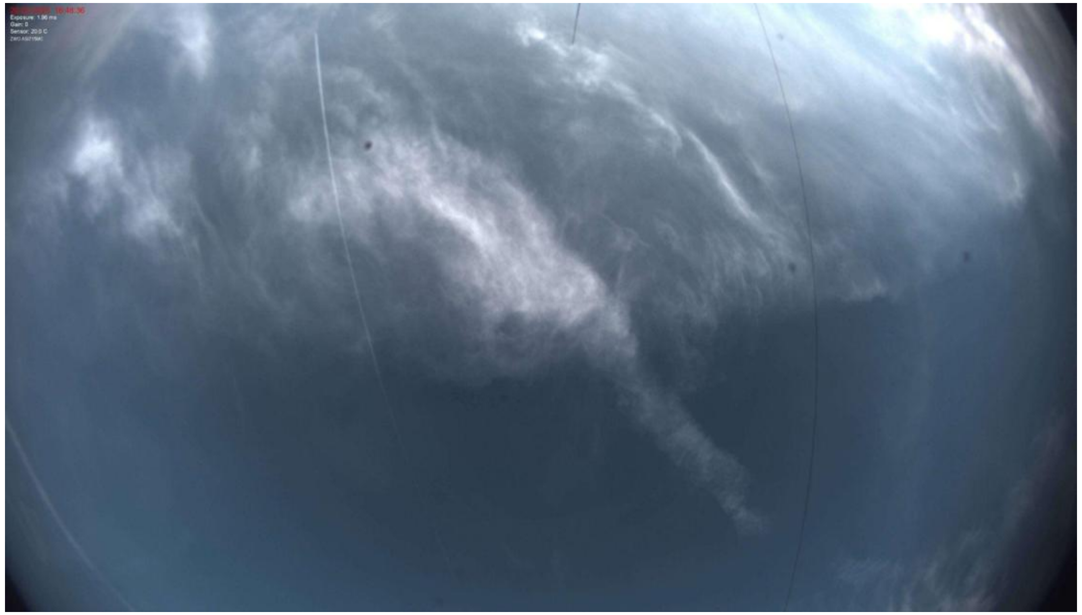
→ Providing **accurate** measurements of **meteorological trends** on the ground and atmosphere

Our Setup

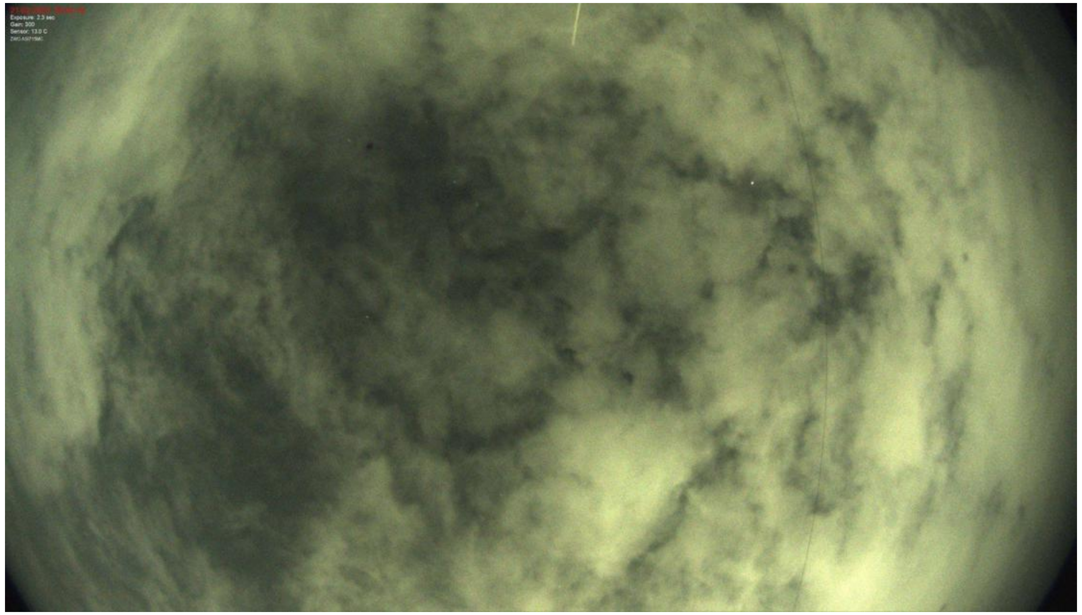
- Core network measures **every 2 minutes**
  - **RTT, Traceroute, Upload / Download** Throughput (UDP)
- Against **static cloud server** in Germany
- **High-resolution ping** latency tests (**every 10ms**) against Starlink POP Gateway
  - Allows precise latency analysis on 1-second basis
- Tests are executed on an openWRT router **directly attached to Starlink**
- Results are stored on a **Raspberry Pi** (offline-capable)
  - Regular upload to storage server for **Live Observation**
  - **Separate** WAN link used to keep Starlink interface unaffected

Measuring Cloud Cover (1/4)

- 50-day study period (February 20, 2025, to April 10, 2025)
- Images were taken every two minutes, resulting in a total of 35,325 images (~22.76 GB).



Daylight Image with few Clouds (3/8)



Night Image with Cloud Coverage (6/8)

Measuring Cloud Cover (2/4)

- Scale of **0-8** is used to measure cloud cover
  - 0 being cloudless
  - 8 being completely covered

- Typically conducted by **analyzing satellite images** or using **radar systems** → lack of high resolution to view the direct section above the dish

- A nearby **ceilometer**, an atmospheric lidar instrument capable of determining **cloud height** and coverage based on backscatter, was used too

Measuring Cloud Cover (3/4)

For classification, we used LLMs: „You are an **expert meteorologist and computer vision analyst**. Analyze the sky image and respond in strict JSON format. Do not add any explanation.“

Tasks:

1. Determine if the image is a valid photo of the sky. (true/false)
2. If not valid, briefly explain why (e.g., no sky).
3. Estimate the cloud amount on a scale from 0 (clear sky) to 8 (fully overcast).
4. Briefly describe any visible lens effects or artifacts and how much percent of the picture is affected by that (e.g., lens flare: 20, dirt: 5, reflection: 50).

[...]

Hints: Please be aware that there are also a lot of clear sky images, which might look invalid at first glance, especially with lens flares. [...]

Measuring Cloud Cover (4/4)

- Iterative training of 300 images:

Prompt Version	Accuracy Score	Avg. Cloud Error	Invalid / Errors
v1 - Basic Instructions	72.99%	±1.54 oktas	25.33%
v2 - With Examples	77.62%	±1.14 oktas	26.67%
v3 - Artifact Handling	80.55%	±1.00 oktas	15.00%
v4 - Final Prompt	81.72%	±0.97 oktas	14.33%
v4 - *Final Prompt (Train Set)*	84.08%	±0.85 oktas	13.55%
v4 - **Final Prompt (Test Set)**	**90.18%**	**±0.61 oktas**	**5.60%**

Table 5.1: Comparison of Prompt Versions by Classification Accuracy and Reliability

- By making a preliminary assessment whether an image is **valid for classification**, we were able to increase accuracy from avg. cloud diff: **0.99 to 0.61** in the test set (500 images).

Detection of Connected Satellites (1/2)

- Since **mid-2021**, it has been impossible to retrieve the satellite connected to the user terminal (UT) of the dish (removed by Starlink)
- In [AZP24], the authors showed the possibility of **using maintenance/diagnostic data** from a gRPC interface (details of the obstruction map)

Listing 4.1: gRPC status command

```
$ grpcurl -plaintext -d {\"get_status\":{}} 192.168.100.1:9200
   SpaceX.API.Device.Device/Handle
```

Listing 4.2: gRPC Get Obstruction Map Command

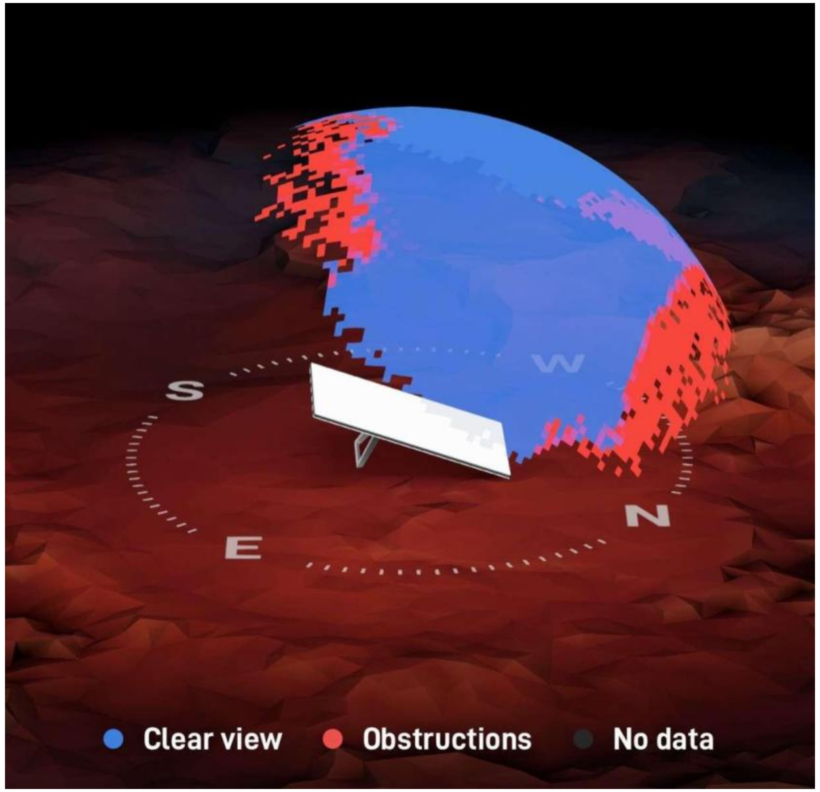
```
$ grpcurl -plaintext -d {\"dish_get_obstruction_map\":{}} 192.168.100.1:9200
   SpaceX.API.Device.Device/Handle
```

Detection of Connected Satellites (2/2)

- Maps the marked pixel to a specific location in the obstruction map / spaces coordinates

- The **Two-Line Element Set Data (TLE)** from **Celestrak** is used to obtain the current positions and names of the connected satellites

- Success of this method was visually validated by the authors in more than **99% of 1000 cases**

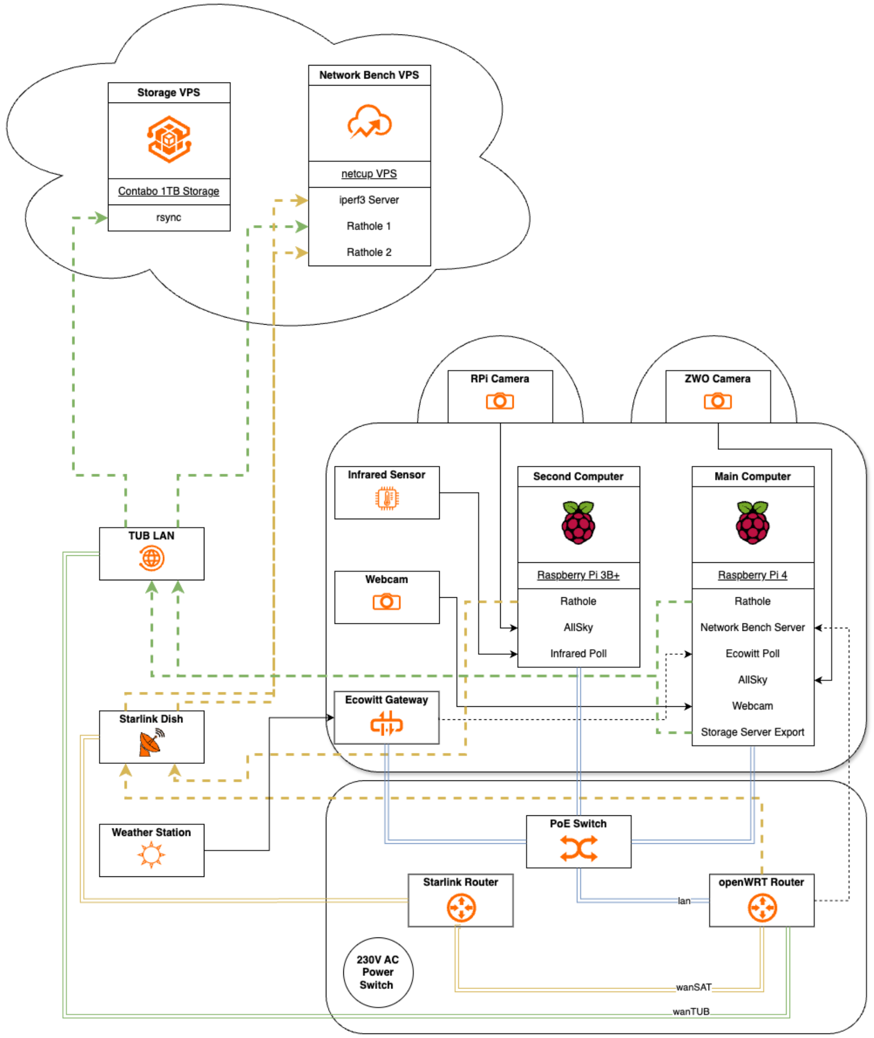


Deployment

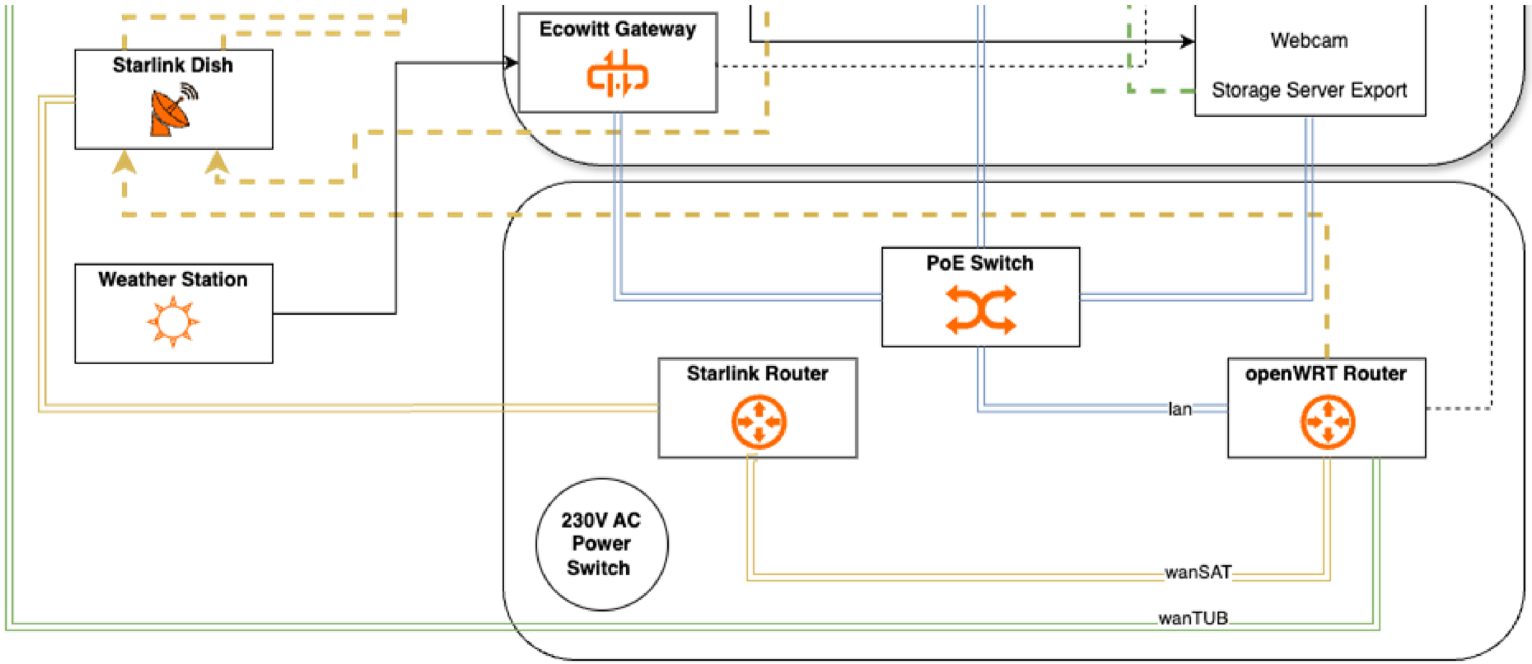
- Deployment on the TU Berlin **main building roof**
  - Place provided by the TUB **amateur radio group**
- Setup requires a modular, waterproof housing withstanding
  - Wind speeds of up to **160 km/h**
  - Low and high temperatures
  - Heavy rain / **moisture**
- Double box setup was protected with **waterproof sealing tape**
- **Weighted** with bricks
- **2-week** testing period

Sealed Box Setup

Hardware Setup



Hardware Setup



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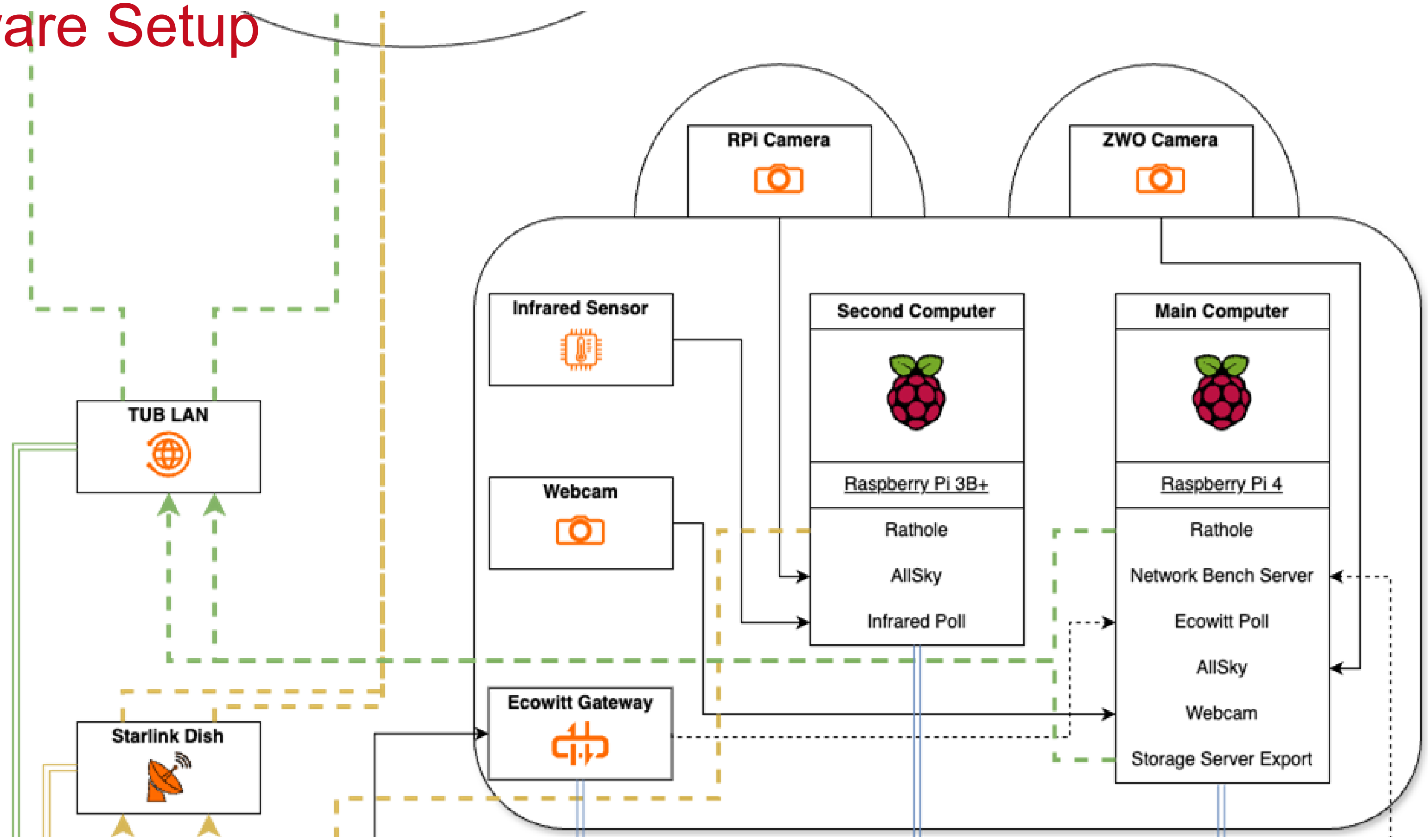

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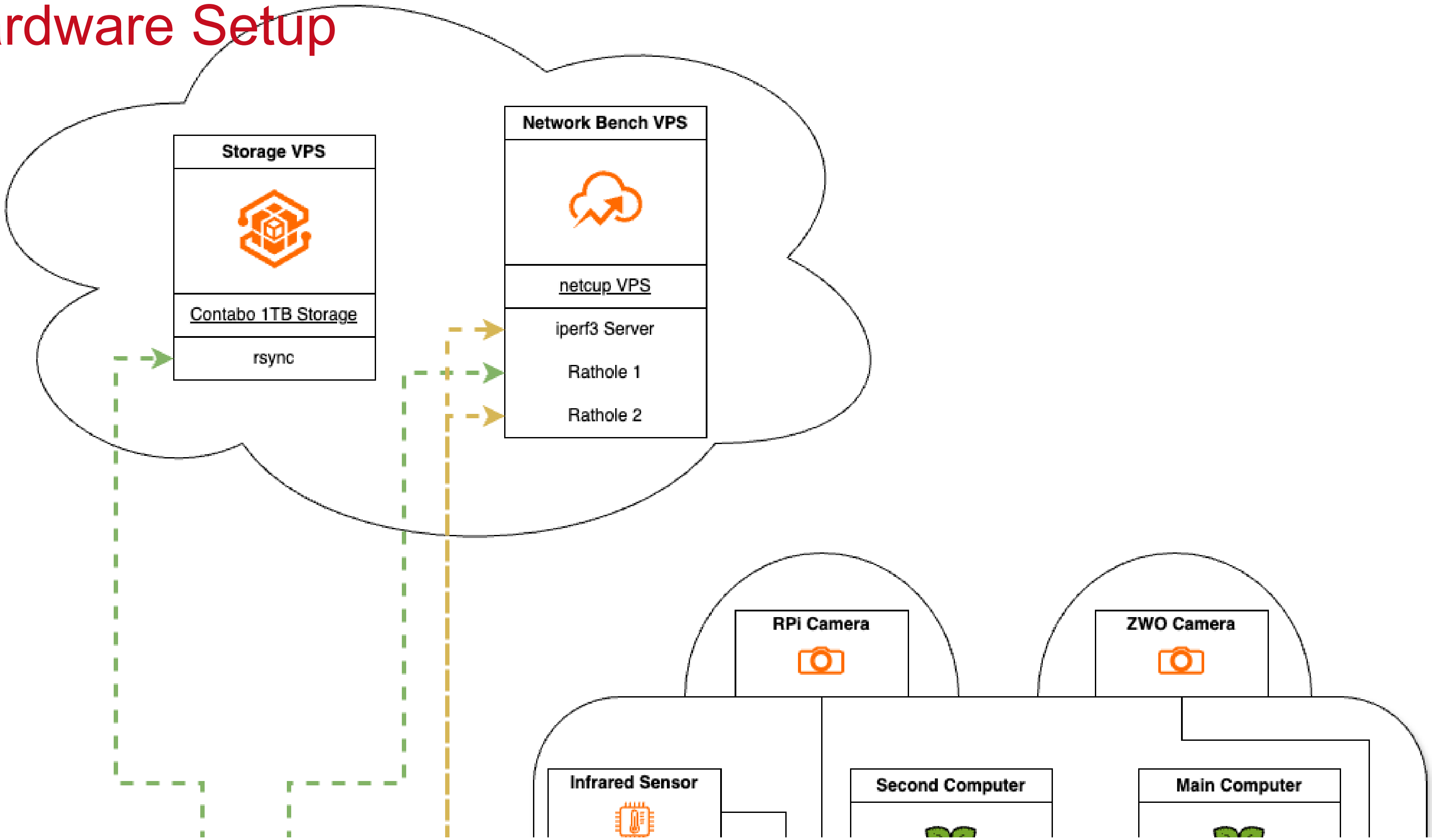
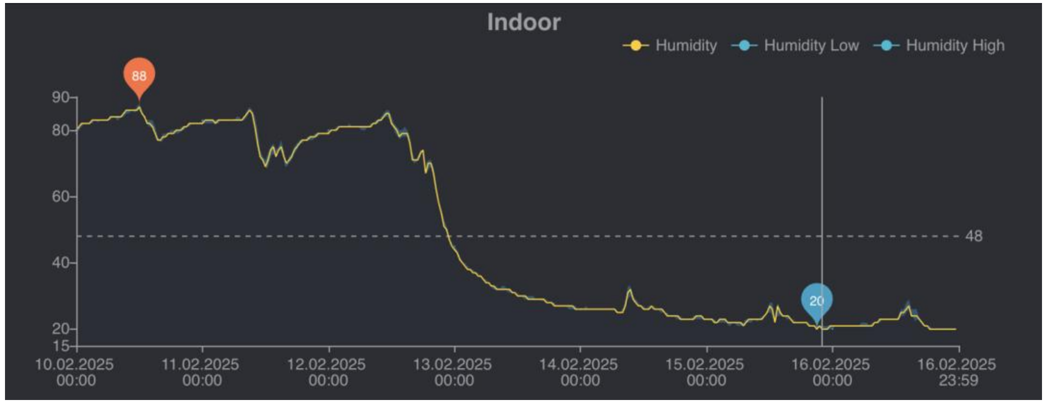


Deployment - Testing Period Adjustments

- Random Router Crashes & Reboot Loops
  - Replacement with a newer and more powerful openWRT router
- Moisture problem in the boxes:
  - Up to **93%** humidity inside
  - Additional **foam sealing tape**
  - 1200g **Silica Gel**
  - 3D printed **sealing clips**
  - Cable holes were **hot-glued**

⇨ **56-day** experiment period



Humidity in the boxes before and after adjustments

Silica Gel

Foam Sealing Tape

Deployment

(a) Stacked Boxes

(b) Starlink, Box Setup and Weather Station

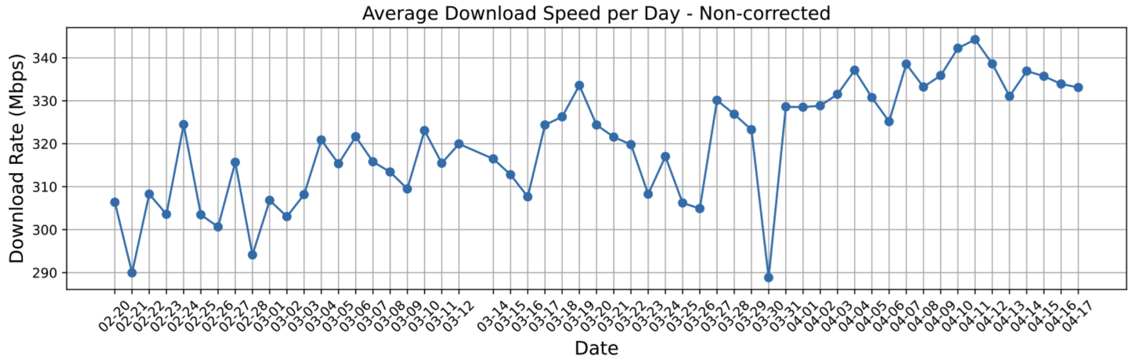
(c) Weather Station

(d) Microwave Radiometer (left), Ceilometer (right)

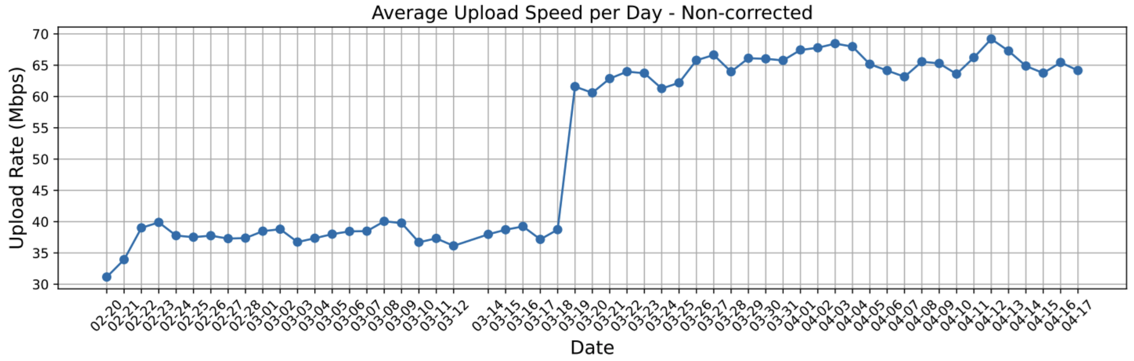
Results - Data Inspection

- Overall **increasing trend** of Upload, Download
- Significant and **abrupt increase** in Upload on March 18th
  - Local changes were ruled out
  - Starlink support confirmed **POP infrastructure changes**
- Overall **noisy connection** in Download, Upload, Ping
- **Hourly drift** in bandwidth (peak at **5am**, lowest at **20pm**)

⇨ **Data Correction** required



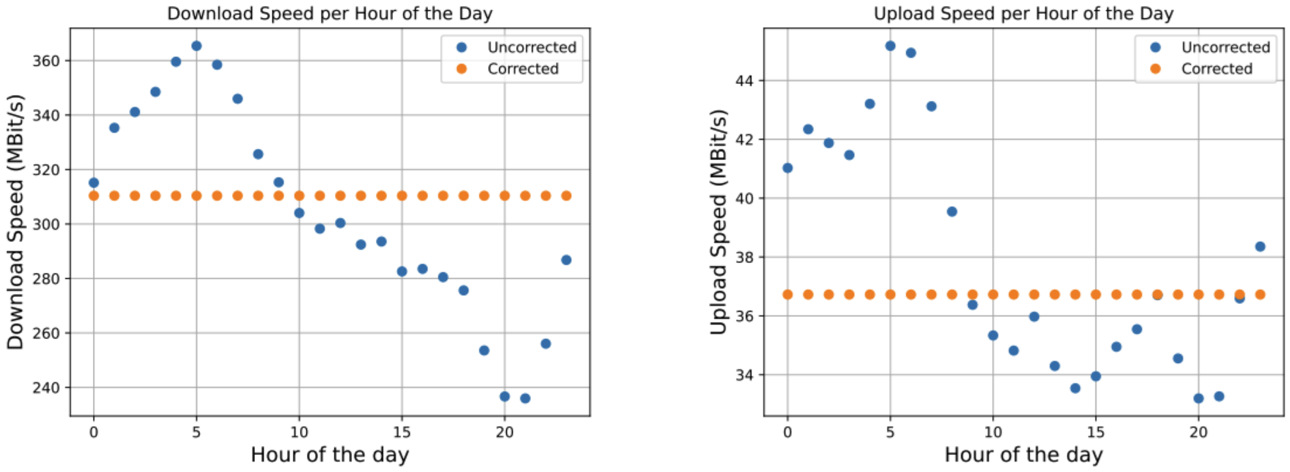
(a) Raw Download Throughput per Day



(b) Raw Upload Throughput per Day

Results - Data Correction

- **Hourly Drift**:
  - Computed **mean throughput** per hour during **good conditions**
  - Difference of **global mean** (all hours) and **hourly local mean** applied as correction
- **Global trend correction** (regression fit)
- **Noise reduction** by **1-hour resample**
  - Overcome **low spatial resolution** of ceilometer and MWR
  - Better representation of weather condition trends



(a) Download Throughput Correction

(b) Upload Throughput Correction

Figure 6.3: Day Drift Correction

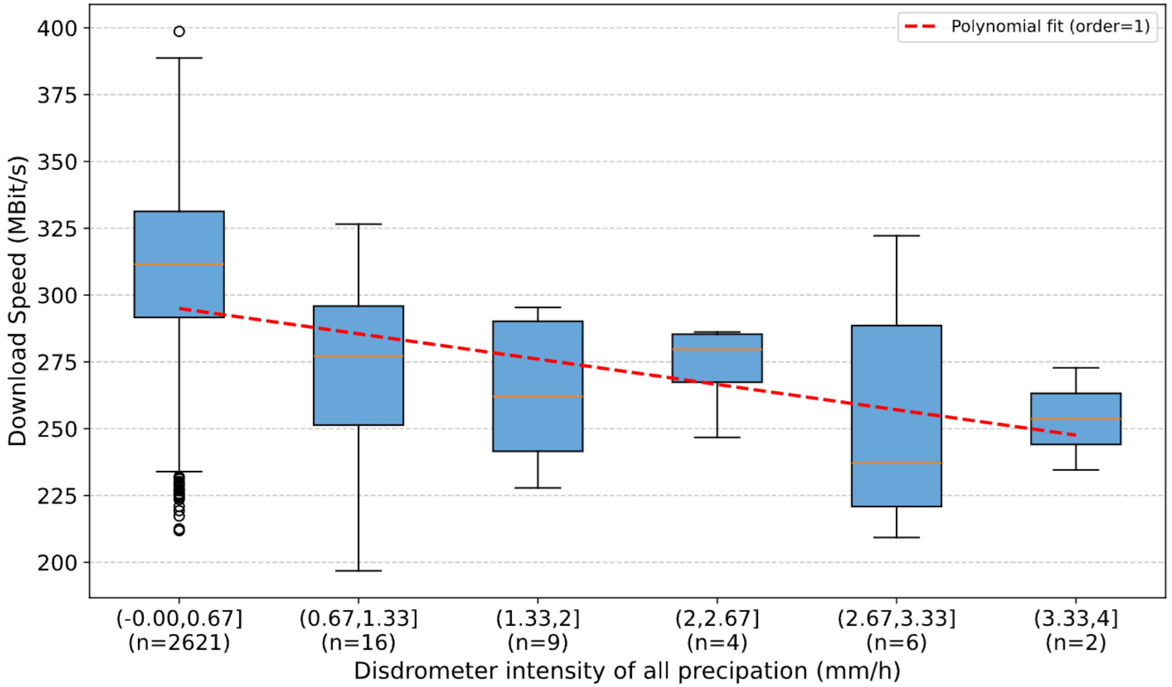
$$\text{drift\_correction}_h = \overline{T}_{\text{all}} - \overline{T}_h \tag{6.1}$$

where $\overline{T}_{\text{all}}$ is the mean throughput across all hours (baseline) and $\overline{T}_h$ is the mean throughput for hour h.

Results – Precipitation (mainly Rain in our case)

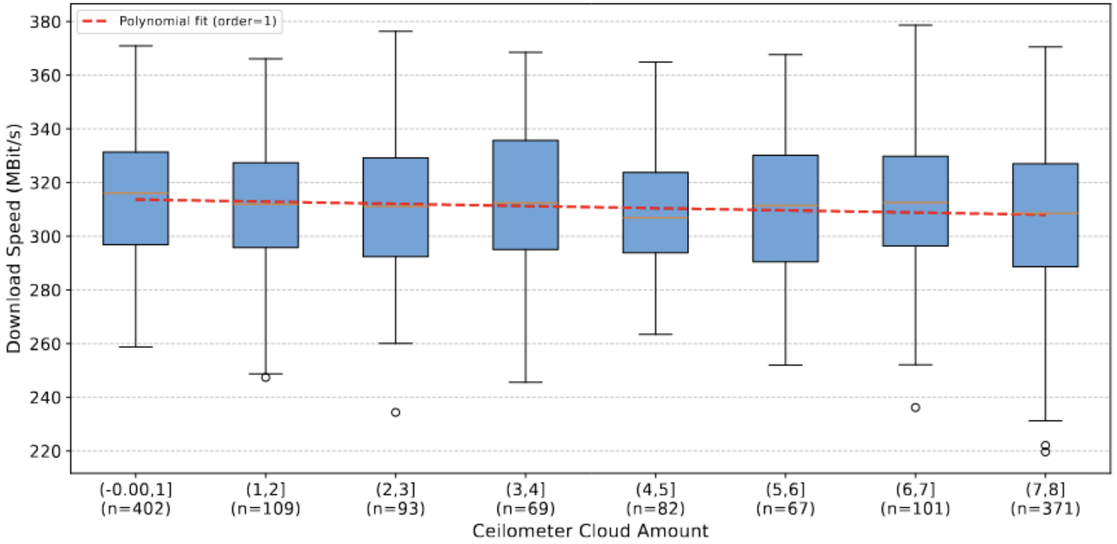
- **Rain** substantial **decreases download** bandwidth - up to **60 MBit/s**
- **Upload** and **RTT** remain largely **unaffected**
- Effects can be explained by the **attenuation** and **scattering** of radio waves caused by **water droplets**

Network Metric	Sample Size	r Value	p Value
Download Speed	1330	-0.25	1.39×10^{-20}
Upload Speed	1330	-0.05	0.070
Average Ping	1330	0.02	0.452



Results - Cloud Cover (1/3 Ceilometer)

- **Cloud cover** measured by Ceilometer shows smaller effect
- Difference of **10 MBit/s in download** when comparing amount **0 to 8**
- Ceilometer measurements only **partially** explain link quality decrease



Network Metric	With Rain			Without Rain		
	Sample Size	r Value	p Value	Sample Size	r Value	p Value
Download Speed	1302	-0.14	2.62×10^{-7}	1288	-0.11	8.34×10^{-5}
Upload Speed	1302	0.01	0.743	1288	0.03	0.344
Average Ping	1302	-0.00	0.906	1288	-0.01	0.617

Results - Cloud Cover (2/3 Image Classifier)

- Image Classifier provides solid estimation for **cloud amount** of ceilometer
  - Mean difference to ceilometer **1.40**
- Similar to ceilometer: **little direct impact**
- Cloud amount only is not sufficient as estimator

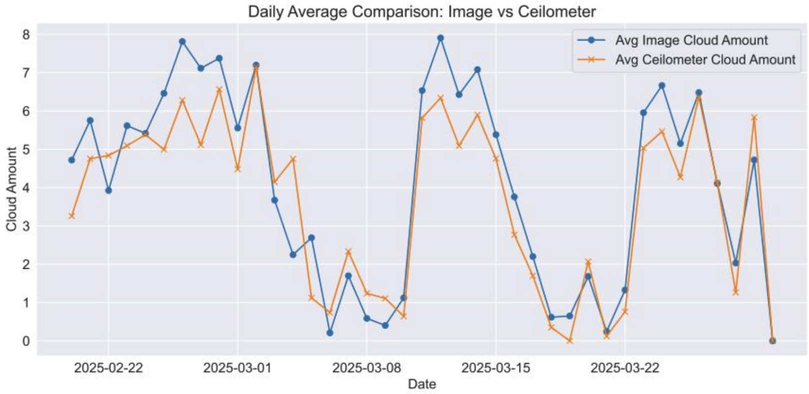


Figure 5.4: Image Classifier vs. Ceilometer - Daily Average Cloud Amount

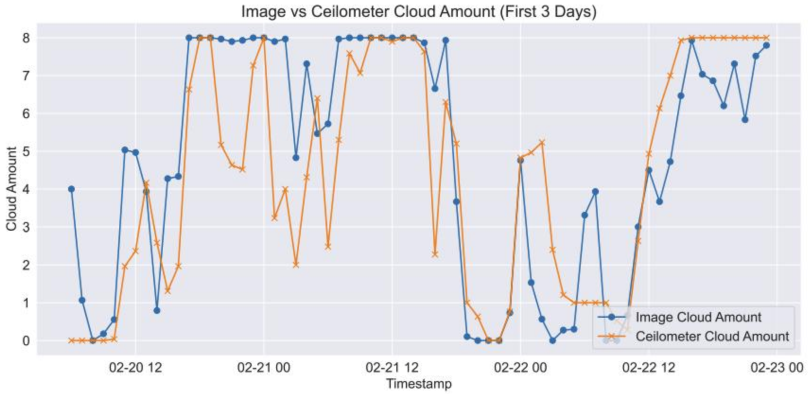


Figure 5.5: Image Classifier vs. Ceilometer - Hourly Average Cloud Amount

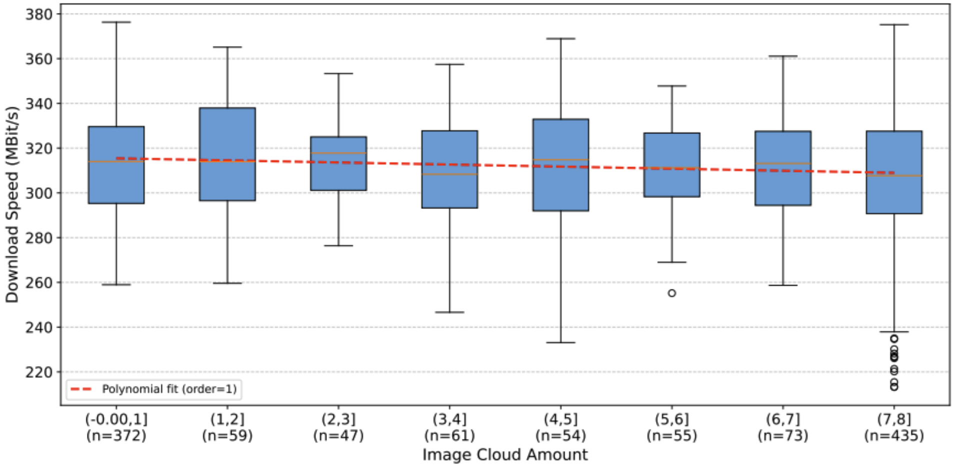


Figure 6.8: Download Speed vs. Image Cloud Amount, Rain Filtered [1h RS]

Results - Cloud Cover (3/3 Liquid Water Path)

- **LWP**: total amount of **liquid water** in a vertical column (g/m2) has **significant impact**
  - Decrease of approx. **30 MBit/s** in download (rain filtered)
- Integrated Water Vapor (IWV): no significant impact
  - vaporized form (gas phase)

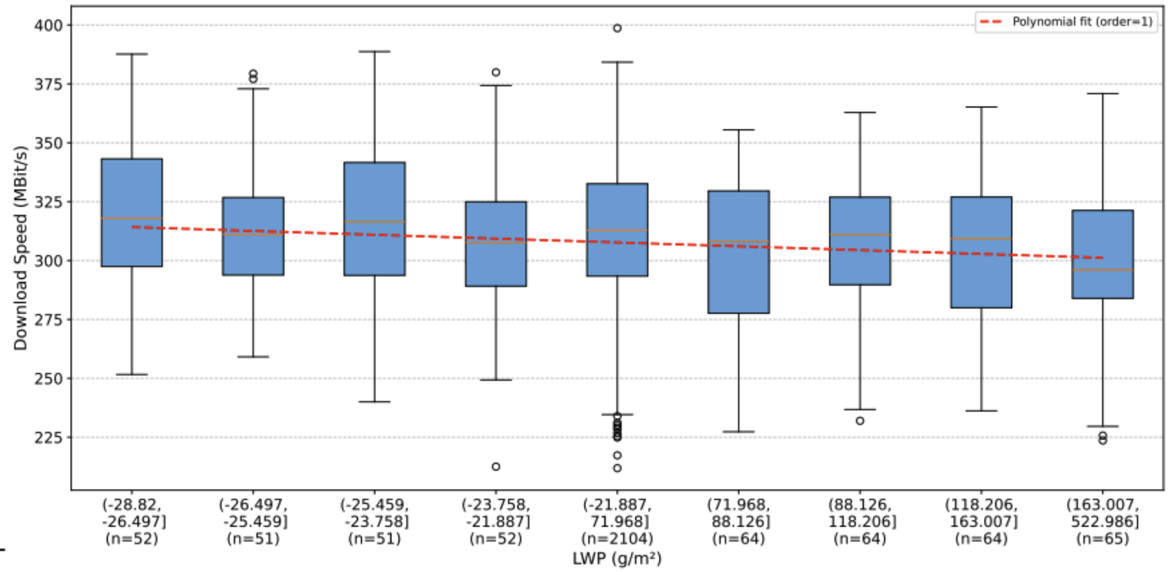


Figure 6.5: Download Speed vs. Liquid Water Path, Rain Filtered [1h RS]

Network Metric	With Rain			Without Rain		
	Sample Size	r Value	p Value	Sample Size	r Value	p Value
Download Speed	1327	-0.28	4.80×10^{-25}	1313	-0.15	5.46×10^{-8}
Upload Speed	1327	-0.06	0.0235	1313	-0.02	0.372
Average Ping	1327	0.07	0.00710	1313	0.08	0.00343

Table 6.9: Correlation of Network Metrics vs. LWP During Rain and No Rain Conditions

Some additional results: re(connections) to LEOs

- **Latency spikes** visible in reconfiguration intervals (seconds 12, 27, 42, 57)
- At each reconf. second, a **new satellite** may be **connected**
  - **Reconnection** to **same** satellite **possible**
- **Latency increases** with **higher consecutive count** and **distance**
- On average a new satellite was connected **every 1.8** intervals (every 27s)

Consecutive Count	Avg. Distance (km)	Avg. Latency (ms)
1	577.27	27.63
2	559.49	27.50
3	560.56	27.51
4	572.35	27.69
5	588.45	28.06
6	607.92	28.51
7	628.68	29.40
8	651.62	29.95
9	691.06	32.12

Table 4.2: Average Distance and Ping Latency grouped by Satellite Consecutive Count

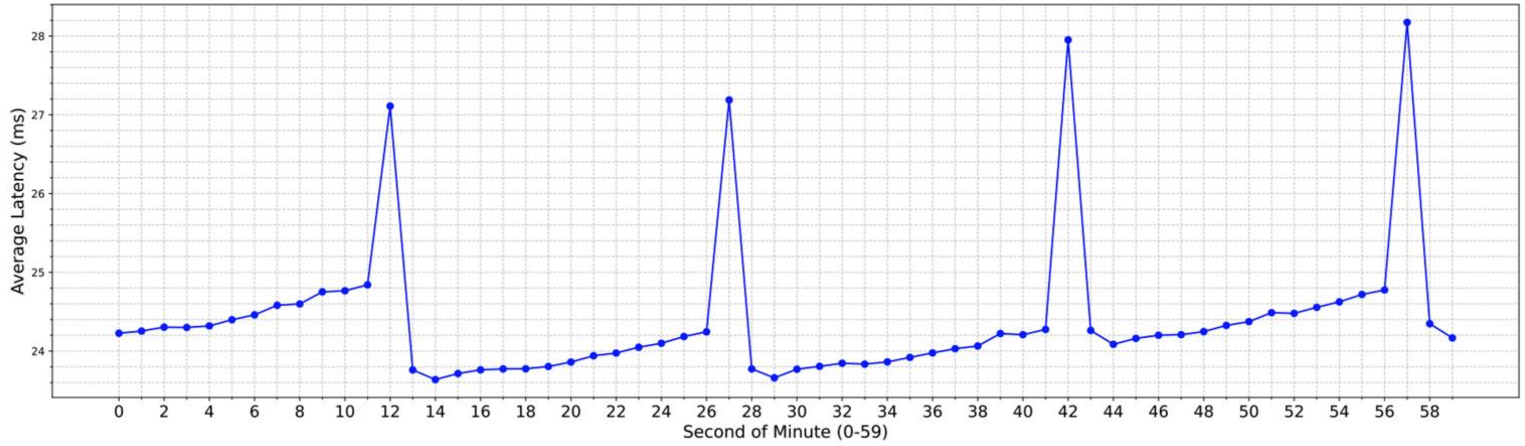


Figure 4.2: Average Latency by Second of Minute

Some additional results: satellite versions

- v1.0 and v1.5 show similar performance
- New Starlink **v2-Mini** Satellite version **increases performance**
  - 34 MBit/s in Download
  - 2 MBit/s in Upload
  - 1.4ms in Latency
- New version in **lower orbit**

Version	Latency (ms)	Download (MBit/s)	Upload (MBit/s)	Distance (km)
Starlink v1.0	25.26	296.38	35.51	610.99
Starlink v1.5	25.29	294.36	35.49	627.79
Starlink v2-Mini	23.82	326.82	38.07	533.27

Table 4.3: Network Performance by Starlink Satellite Version

Discussion / Future Work

- Sensitive to **atmospheric liquid water** rather than to overall cloud presence or water vapor
- Rain events leading to app. **performance degradation**, predominantly in the downlink
- Quality of service is modulated by dynamic satellite, handovers and ongoing technological upgrades in the constellation

Possible Future Adaptations:

- Extend our study in time and space, explore other seasons and continents as well
- Effect of additional kinds of precipitation, such as snow or hail

Thank you for your attention!

Questions?